Efficiency and acceleration in the accumulation of children's early vocabulary

Children's early vocabulary grows very rapidly during early childhood, with large variation between children, and one important view is that this growth can be characterized as a process of accumulation (McMurray, 2007; Mitchell & McMurray, 2009; Hidaka, 2013; Mollica & Piantadosi, 2017; Kachergis et al., 2022). Each distinct word can be conceptualized as a "bucket," and every token of language input then is a "drop" that falls into its respective bucket. When each bucket is full, the word is learned. The accumulator view has been especially influential because it connects with a viewpoint in which the quantity of language that children hear plays a causal role in the speed of their learning (Fernald & Marchman, 2012). But despite the promise of accumulator models, they have largely not been used to understand variation between individuals. Further, the rapidity of children's early word learning stands as a rebuke to a naive accumulator view. Sometimes exposure to a small number of examples of a word in the lab is enough for a child to form a lasting representation. We address these puzzles through systematic investigation of accumulator models.

We compared a pure accumulator (model 1) to an accelerating accumulator (model 2), in which the rate of accumulation increases with development. Model 3 adds variation in efficiency (accumulation rate; the model intercept); model 4 adds variation in acceleration (the slope). These extensions define longitudinal growth models that can be fit directly to large-scale datasets from the MacArthur-Bates Communicative Development Inventories (CDI), including longitudinal and cross-sectional data from Wordbank (Frank et al., 2017). Across all datasets, model 4 provided the best fit, with estimated acceleration vastly exceeding pure accumulator models (Figure 1).

The two model parameters (efficiency and acceleration) provide a principled decomposition of factors influencing growth. Applied to cross-sectional and longitudinal data (24 and 3 languages, respectively), parameter estimates suggested that the female advantage in early vocabulary results primarily from differences in overall learning efficiency – stable sex-related variation with limited compounding over time. In contrast, socioeconomic differences from maternal education appear in both efficiency and acceleration and may compound across development.

We next estimated the contributions to vocabulary growth of 1) children's language processing (measured using the looking-while-listening task) and 2) their language input environment (measured using LENA and headcams). We used four datasets containing both CDI outcome data and language exposure data, with four having processing data as well (Egan-Dailey & Bergelson, 2025; Fernald & Marchman, 2012; Fernald, Marchman, & Weisleder, 2013; Adams et al., 2018). The language environment in these datasets was related to variation in acceleration – suggesting compounding effects over time – while processing was primarily related to efficiency, not acceleration (Figure 2).

Overall, our analysis supports a view in which children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. This model describes how learning efficiency for individual words increases over orders of magnitude across childhood. Further, the parameters of the model provide a useful lever for understanding individual and group-level variation.

(493 words)

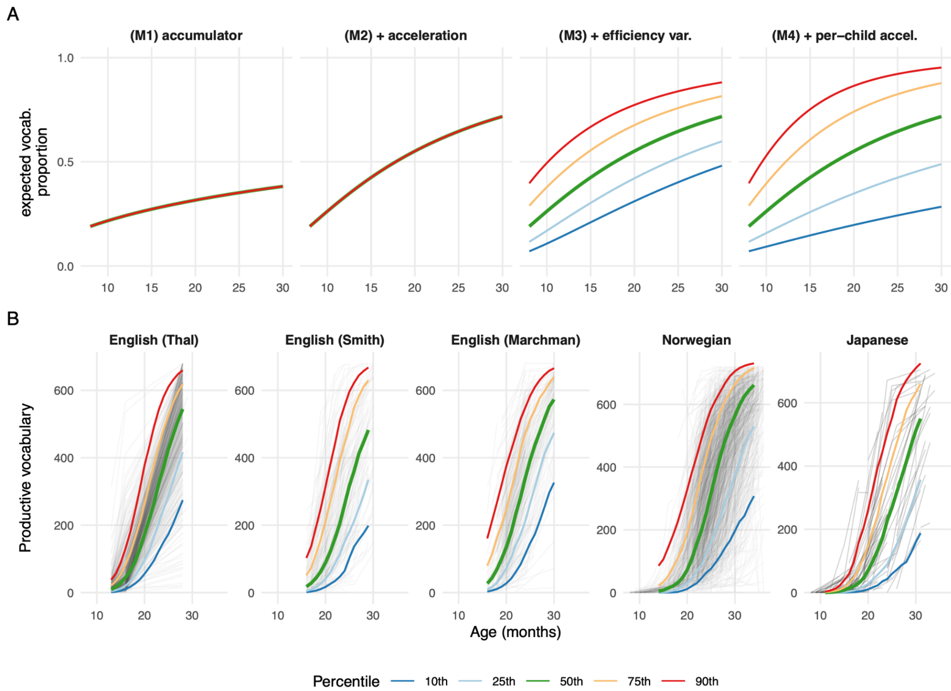

Figure 1. Top: Model variants showing implied vocabulary growth curves. Bottom: Model 4 (accelerating accumulator), fitted to longitudinal data from five datasets from Wordbank. ΔAIC for Model 4 was > 1000 in all cases. Colored lines show quantiles.

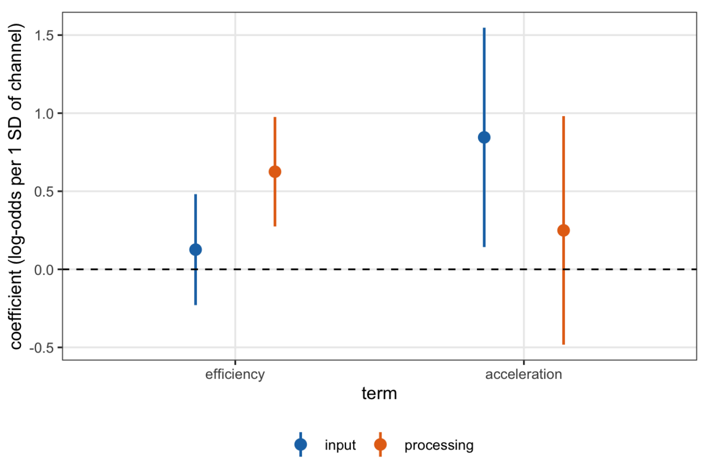

Figure 2. Coefficient estimates for input and processing channels in the accelerating accumulator model. Input predictor is the standardized average words per hour from home recordings (LENA or BabyView camera); processing predictor is residual age-standardized looking-while-listening speed. Error bars show 95% confidence intervals.

References

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Children's early vocabulary grows very rapidly during early childhood, with large variation between children, and one important view is that this growth can be characterized as a process of accumulation. But despite the promise of accumulator models, they have largely not been used to understand variation between individuals. We fit a series of accumulator models to longitudinal CDI data with home language environment and processing data as predictors. We find that children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. Our best-fitting model describes how learning efficiency for individual words increases over orders of magnitude across childhood. Further, the parameters of the model provide a useful lever for understanding individual and group-level variation. In particular, we find evidence that input environment effects compound over time, while processing measures index stable differences in efficiency.